

Understanding Uses' Needs in the Development of Indoor Navigation for Visually Impaired Users in Saudi Arabia

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ABSTRACT

People with visual impairments find it challenging to navigate unfamiliar buildings independently. This study aimed to understand the experiences and attitudes of people with visual impairments in Saudi Arabia toward indoor assistive technologies and the impact of using such tools. To this end, we conducted an experiment that involved interviewing and empirically comparing the performance and perceived mental workload of people with visual impairments in Saudi Arabia using the two most-used technologies for indoor navigation. The results indicate that Saudi users with visual impairments rely on sighted guides to navigate confidently inside unfamiliar buildings rather than on technologies. This reliance affects their trust in and use of technical assistive technology. Based on their attitudes and experiences, this study provides several design recommendations that can be used as a guide for building assistive tools for Saudis and other people with visual impairments.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility design and evaluation methods; Accessibility systems and tools.**

KEYWORDS

Assistive technology; interaction analysis; interdependence; vision impairment; sighted guiding.

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1 INTRODUCTION

People with visual impairments face difficulties in walking confidently inside unfamiliar buildings [21]. To mitigate these challenges, many assistive technologies (known as Electronic Travel Aids [21]) have been proposed. Examples of these technologies are smartphone applications [3, 27], wearable systems [36], smart suitcases [23], electronic cane [9, 25], and assistive robot [7, 24]. However,

few assistive technologies have been developed or employed in Arabic [5], especially for Saudi users. According to the latest publicly available report from the General Authority of Statistics in Saudi Arabia [12], 2.9% of people in that country have disabilities, with the highest percentage (46.02%) with visual impairments.

This research aimed to understand how individuals with visual impairments in Saudi Arabia navigate indoors, what assistive technologies they use to navigate independently, and how they feel about these technologies. To achieve this aim, we performed an experiment that has three main phases. First, we conducted a pre-experiment interview with five blind people to understand their needs and the challenges they face when navigating indoor spaces and to determine what assistive tools they use to navigate unfamiliar buildings. Secondly, we conducted an empirical experiment that compared participants' performance when using the two most popular navigation assistive technologies. Finally, we conducted semi-structured interviews with the participants to gain further insight into their impressions of the technologies they used and the specific design details of these technologies. Our findings improved our understanding of blind people's needs and attitudes, as well as the relationship between assistive technology and the perceived workload of these tools.

2 RELATED WORK

Several studies have been conducted to understand blind people's behaviors and challenges in indoor navigation [4, 14, 15, 18, 18, 34, 35]. Others aimed to propose new technical assistive technologies and/or measure the impact of the technologies on moving indoors independently [1–3, 6, 11, 13, 17, 23, 26, 31–33]. These technologies can be mainly grouped into wearable, handheld, and mixed reality [28, 34]. Here, we mention some of the most relevant studies.

Guerreiro et al. [17] conducted a study that aimed to understand how turn-by-turn navigation instructions and point-of-interest (POI) location/landmarks could help people with visual impairments create memorable sequential representations. To this end, the authors built and tested two smartphone-based virtual navigation interfaces. These apps mainly support the creation of a mental representation (cognitive map) of the environment. The authors experimented with the use of these apps on 14 visual imprints, reporting that virtual navigation enabled blind users to build accurate sequential representations of the environment. However, the main challenge the participants faced was estimating the length of a street.

Ahmetovic et al. [3] created NavCog, a smartphone-based indoor navigation assistant. This application uses a network of beacons that are pre-deployed in the environment to provide accurate localization. The authors tested the system on six blind people, finding

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that the system could successfully guide visually impaired users in unfamiliar environments. Sato et al. [35] tested a modified version of this application (NavCog3) in a larger environment (a shopping mall) on 43 participants with visual impairments and found that its use increased users' confidence in walking independently in unfamiliar places.

Guerreiro et al. [18] used NavCog to investigate user behavior during indoor navigation and provided recommendations for developing navigation devices. Similarly, Ahmetovic et al. [4] employed NavCog to investigate the types of instructions and information required to support blind people who were using a cane or guide dog for assistance. Their results revealed the need for a personalization option to keep the participants engaged.

Kayukawa et al. [23] proposed an assistive suitcase system (BBeep) to support visually impaired travelers. The system uses preemptive sound notifications to help clear a path by alerting both the user and nearby people about the potential risk of collision. The system was tested by six blind users at an international airport, and it was found to reduce the number of collisions.

May et al. [31] developed and tested a soundscape system that produces sound reflecting off different surfaces to improve accessibility in a virtual reality environment (routing paths or finding objects). This was done by conducting two experiments: one with five participants to understand the requirements for such a system, and the other with three participants to evaluate the system in a real-world building. The authors found that soundscape and spotlight (e.g., object detection) systems can help people with visual impairments quickly map unfamiliar spaces.

3 PRE-EXPERIMENT INTERVIEWS

The purpose of this qualitative study is to understand the needs and challenges that visually impaired individuals in Saudi Arabia face when navigating indoor spaces, as well as the assistive tools they are currently using to navigate unfamiliar buildings. The ethical approval to conduct this study was obtained from Institutional Review Board (IRB).

3.1 Procedure

This study comprised a series of interviews with students with visual impairments in Saudi Arabia. All our participants were either legally or totally blind. We conducted semi-structured interviews with five participants from different educational backgrounds after they signed a consent form. We accommodated the participants' needs by conducting the interviews either in person or remotely.

We started with demographic questions, shown in Table 1. Then, we asked the participants about their experiences in navigating unfamiliar buildings, the tools they are currently using to do so, and the effectiveness of these tools (if any). The interviews were recorded and transcribed for further analysis.

After transcribing the recorded interviews, we used a narrative analysis approach to highlight important aspects of our participants' stories that would best resonate with our readers or highlight the critical points we found in comparative research.

The small sample size resulted from difficulties encountered during the recruitment process. For example, we conducted our study in a very conservative society with a small available pool of

participants. Notably, similar studies have faced the same issues (e.g., [29]). The only available pool for us was computer-literate female students. Difficulties in recruitment are a well-known issue in the studied domain [8, 10]. Some studies have chosen to include sighted participants to increase the sample size and improve statistical significance [30], but we chose to avoid this because sighted participants' performance may not be representative of the performance of users' with visual impairments [20].

3.2 Results

All the participants agreed that they are not using any tool to navigate unfamiliar indoor buildings without in person assistance. They all require a sighted guide, who is usually someone they know (i.e., a family member or friend), holding their hand, navigating the building, and describing the main paths and the directions of important places (e.g., classrooms, restrooms, etc.). "Who sighted guide is going to join me?" is a question the participants are often worried about. Three participants said there was no way to go to an unfamiliar building alone, even if the reason to go there was important. One participant mentioned that "however the technology and tools have improved, they will never be like a real person." Two participants had different alternatives, if needed: 1) asking a stranger in the front of the building (P5) to guide them and 2) using a white cane or a navigation-assistant mobile-based application (BeMyEyes) (P1). P1, who was fully blind, hired an assistant to navigate around campus.

After the first visit with the sighted guide, the participants would memorize a map of a building "super-fast." Thus, for the next visit, they could navigate independently with assistive tools (see Table 1). They explained the map when we asked them the following: "Can you describe the map that you memorized and how to use it to go to a specific place (i.e., your classroom)?" All described it from the main gate of the building, what is left and right, and which path leads to the place they need. We noticed that they all mentioned stairs, columns, and slopes in their maps.

After memorizing the map, assistive tools can be used, such as a cane and tactile paving surfaces. None of the five participants could move independently with no tools, regardless of how many times they visited the building ("At least [there is a] tactile paving surface. It [would] be hard to navigate without it.").

During the interview, the participants described their feelings when navigating unfamiliar indoor buildings by using the words "bad," "loss," and "strange," followed by deep sighs. P5 said, "My lifelong problem is mov[ing] from one room to another; I cannot do it." Three of the participants remembered having many accidents, such as falling on stairs and hitting objects and walls.

To understand the source of their fears and challenges, we asked if they usually ask someone how to navigate a new building before going. They all had asked about stairs and floors (flat or with slopes). Those who were partially blind asked about the light not being too bright or too dark, the columns, and objects (e.g., tables, chairs ...etc).

All the participants were aware of at least one existing assistive tool. We categorized those tools as self-navigation applications with maps (e.g., IBM NavCog), volunteer-guided applications (e.g., BeMyEyes), and image recognition applications (e.g., EnVision). Only

Table 1: Descriptive Information for the Participants

ID	Age	vision impairment	Education	Navigation Aids
P1	31	Totally blind	Postgraduate	Sighted guide(primarily), White cane, Tactile paving surfaces, Braille signs
P2	25	Legally blind	Postgraduate	Sighted guide (primarily), Ask others, Tactile paving surfaces
P3	22	Legally blind	Undergraduate	Sighted guide (primarily) , White cane, Tactile paving surfaces
P4	22	Legally blind	Undergraduate	Sighted guide (primarily), White cane
P5	23	Legally blind	Undergraduate	Sighted guide (primarily), White cane

two participants were currently using a tool (mobile applications): P1 uses BeMyEyes, and P5 uses EnVision. The former mainly helps navigate indoors, and the latter recognizes objects and faces around the user.

To further understand the limitations and the possibility of developing effective assistive tools, we conducted the following experiment, which experimentally compared their evaluations of using two different navigation tools.

4 EXPERIMENT

We conducted this experiment to investigate the impact of using self-navigation assistance tools in guiding Saudi people with visual impairments in unfamiliar environments. In this experiment, we asked the same participants of the first study (Table 1) to attend a follow-up research experiment; four of the five (P2–P5) participated in this experiment. This experiment primarily evaluated the performance of using two of the most used navigating tools: NavCog (a fully automated application) and BeMyEyes (a volunteer-based application). The reason behind choosing these tools is the participants' familiarity with them as well as these tool are widely used by the population [5]. For example, the BeMyEyes were the only tool that is currently used by the participants. Our participants have not used NavCog directly, but all our participants have mentioned Google Maps which has a similar functionality (i.e., turn-by-turn navigation). However, Google Maps is unavailable in all buildings and cannot be modified easily (i.e., not an open source), while NavCog is open source and can be deployed and modified easily.

4.1 Environment and Apparatus

We conducted the experiment inside a building of the College of Computer and Information Sciences at King Saud University, after obtaining ethical approval from the Institutional Review Board (IRB) to protect the rights and welfare of our participants. The building was unfamiliar to all participants, as they were all non-computer science students (law, translations, and social services). In this building, we identified six paths, the total length of all of which was 200 m. Figure 1 shows the floor map, locations, and paths.

In this experiment, we used a within-subject design with two conditions. These conditions are two navigation assistance tools: NavCog and BeMyEyes, which have been modified or imitated for easy use by our participants.

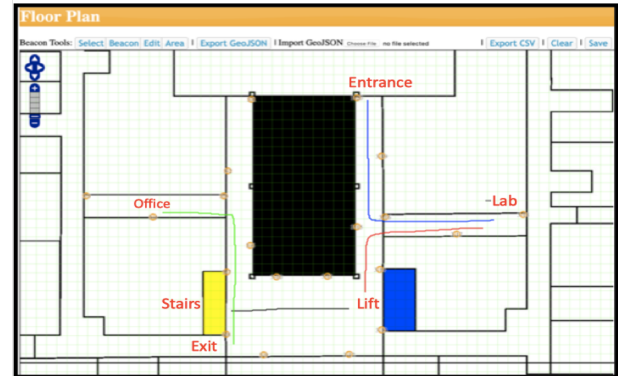
NavCog: This app relies on Bluetooth Low Energy (BLE) beacons and machine learning (KNN) to provide accurate localization

and activity recognition. This app is primarily used to help people navigate indoors using turn-by-turn instructions, and it provides recovery instructions if the users leave the building or planned route. One of the main features of NavCog is the voice interface, which allows users to interact through spoken commands.

Using NavCog effectively in our experiment requires annotating the map, training the model, and, more importantly, updating the interfaces to support the Arabic language. This has been done by adding an Arabic localization file to the application. In the area we chose, we placed 20 beacons at a distance of 7 m to ensure an accurate level of localization (see Figure 1).

BeMyEyes: This mobile application connects visually impaired people to sighted volunteers via live video calls, where this communication allows visually impaired people to know what they are looking at by turning their cameras and asking the volunteers to describe what they see, whether a product expiration date or a new location needing navigation.

To ensure the quality of the experiment (i.e., quality and availability of volunteers), we mimic BeMyEyes by conducting video calls with volunteers using another video calls application. All volunteers were unfamiliar with the buildings. We explained our experiment and what they should do before calling.

**Figure 1: Floor Map, paths, beacons locations**

4.2 Task and Procedure

Each session began with a few questions about their familiarity with the technologies used in the experiment: touchscreen phone, GPS system, and voice-based engine. All four participants use a

smartphone (iPhone) daily, voice-search-based frequently, and GPS systems in the past.

Next, we asked them to perform two tasks, each of which included use of one of the mobile applications to navigate two paths independently chosen randomly from a set of six paths. The two paths in each task differed from each other, as well as from the two paths in the other task. This is to avoid learnability that may be gained by completing a previous task.

During each task, three experimenters followed participants closely to ensure their safety without interfering in the navigation task. One recorded the audio, while the others took notes for analysis purposes.

After completing each task, we asked them a few questions, including giving the NASA-Task Load Index (NASA-TLX) questionnaire about the application used in the task.

4.3 Results

Our main goals were to analyze the impact of navigation apps to support the independent mobility of visually impaired individuals in unfamiliar buildings. We report user performance based on completion time, errors, and perceived mental workload. We used a t-test with an alpha value of 0.05 for all quantitative analyses.

Navigation Time: Participants completed the task of each tool, on average, in 5.9 minutes (SD = 1.9) and 5.3 minutes (SD = 1.3), respectively (Figure 3). The average walking speed of the participants using NavCog was 0.58 m/s (meter per second) (SD = 0.23), and for BeMyEyes, it was 0.53 m/s (SD = 0.11). BeMyEyes tool has a slight advantage, but the differences were not statistically significant (see Figure 2).

Navigation Accuracy: The total number of errors that occurred using NavCog and BeMyEyes was 31 and 7, respectively, with significant differences ($t(7) = 5.01, p < 0.05$). Figure 2 analyses these errors, which were classified as minor and major based on their cause and impact.

Minor errors: These errors are self-recovered errors that do not prevent the participant from completing their path. Examples of minor errors include unclear instructions (design issues), such as misunderstanding the instruction provided, missing a right turn, and technical issues, including recognizing noise around the participant. NavCog presented significantly more errors than BeMyEyes ($t(7) = 4.58, p < 0.05$).

Major errors: These errors prevent participants from successfully completing the path, requiring our intervention to avoid serious injuries or leading to a destination that cannot be easily or quickly identified. BeMyEyes has more major errors.

Perceived Workload The averages of the NASA TLX score (1-100) were *high* for both NavCog (61.67) and BeMyEyes (52.60), according to Hancock and Meshkati's [19] interpretation. Figure 3 shows the mean of NASA-TLX scores for each condition (BeMyEyes and NavCog). The figure shows that BeMyEyes has fewer temporal, physical, frustration, performance, and effort demands, while NavCog has only fewer mental demands. The only significant difference was in temporal demand ($t(3) = 7.84, p < 0.05$).

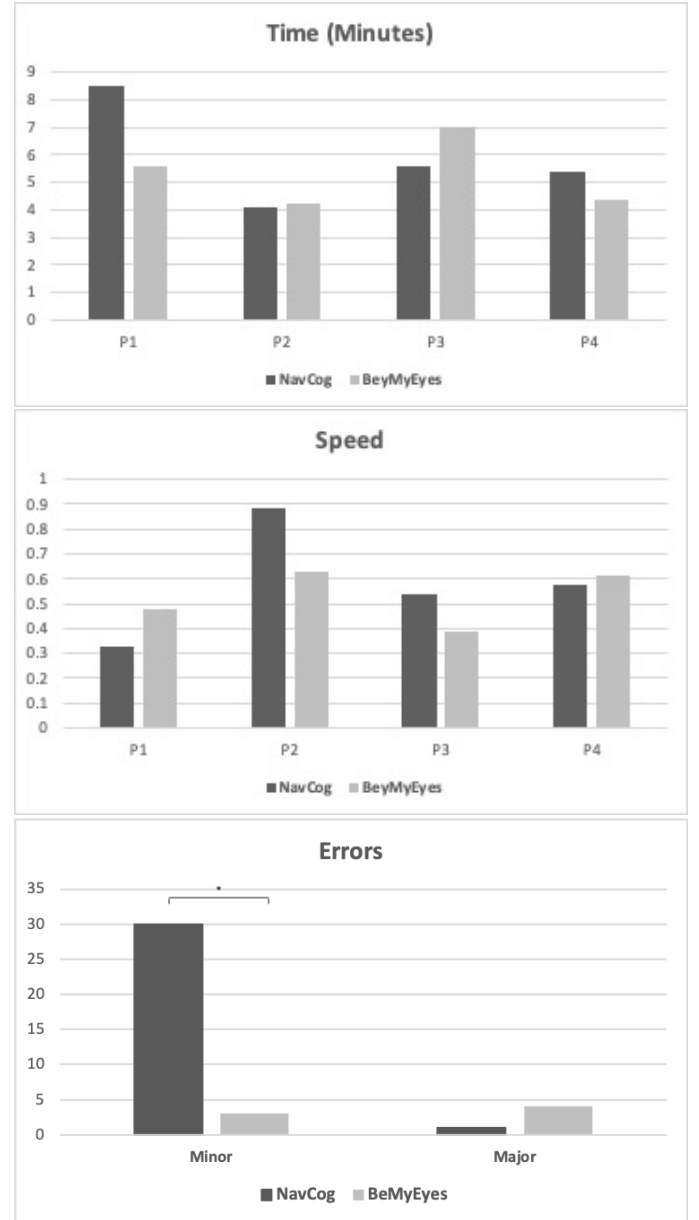


Figure 2: Study Results

5 POST-EXPERIMENT INTERVIEWS

After completing the empirical experiment, we conducted a longer interview with the participants about their overall impressions and the specific design details of the tools. We also asked the participants to tell us which tool would be useful or enjoyable to them.

The participants' comments during the semi-structured interviews were transcribed, coded, and analyzed by two researchers. We performed an in-depth analysis of these post-interviews. The following are the key points on which the participants agreed on.

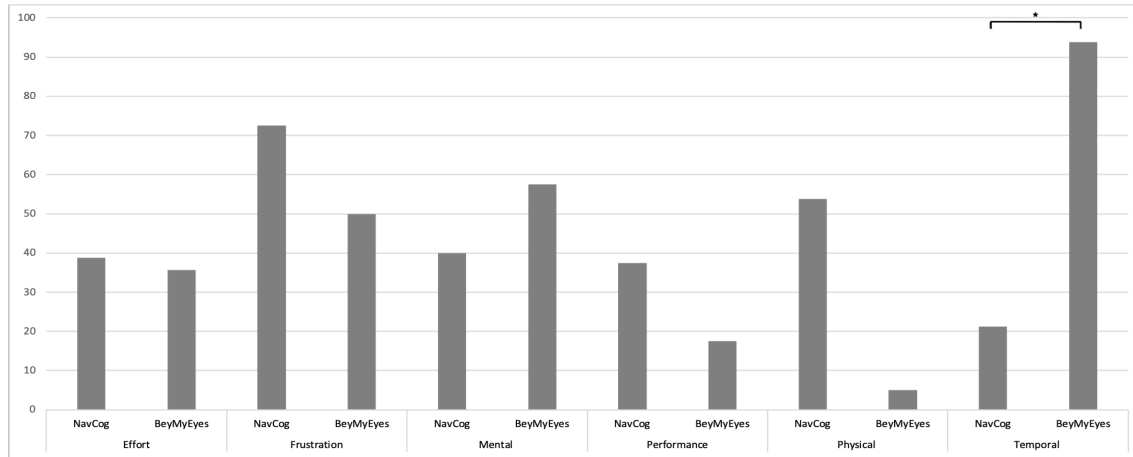


Figure 3: The mean of NASA-TLX scores

5.1 Navigation Instructions

A 5-point Likert scale (strongly disagree=1, disagree=2, neutral=3, agree=4, strongly agree=5) was used to measure the usability of NavCog's navigation instructions. All participants rated the navigation instructions of NavCog unfavorably ($M = 2.24$, $SD = 0.82$), including the inappropriate use of the number of meters left in navigation ($M = 1.5$, $SD = 0.86$) and the frequency of instructions ($M = 2.25$, $SD = 1.6$), which were only provided once. They provided suggestions on how to improve NavCog's navigation instructions by 1) measuring the path using steps instead of meters, 2) reporting the instructions frequently (P5: "at least when I stop"), and 3) detecting objects and slopes in their paths. A participant said "should not ask me to move forward if there is an object ahead."

However, all participants agreed on the accuracy and understandability of the instructions provided by BeMyEyes. The instructions provided by the volunteers used steps to measure distances (e.g., "about five steps in front"). In addition, the two-ways communication allows the participant to confirm and obtain more details. One participant stated that "the volunteer can understand and provide accurate info even if the place is unfamiliar to them."

5.2 Pros and Cons

The participants indicated that the main advantage of NavCog is the independence it provides (two participants stated this clearly). Another advantage is the quality of the search service ($M = 4.753$, $SD = 0.4$, using a 5-point Likert scale). However, it is too slow and cannot guide users with correct directions (e.g., "move in front or in the back"). In addition, it cannot recognize objects and slopes.

The main advantage of BeMyEyes is it's trustworthy; three out of five participants said that they trusted the application. One of them stated that "the one providing the instructions is a volunteer, who should be a nice person." However, participants raised many issues, such as being uncomfortable using the system, as it required them to hold their phone in front of their faces at all times (P2: "it is embarrassing"), and cultural limitations; P3: stated that "if a volunteer is a man, I cannot navigate inside women-only spaces."

The third point raised by P3 was unstable, as it relied on the internet to ensure the quality of the video and the safety of a route.

Overall, participants liked using the assistive applications ($M = 3.66$, $SD = 0.8$). Two participants preferred NavCog ("new technology and idea" and "I like everything in this app"), and two chose to use BeMyEyes ("accurate navigation" and "quick response").

6 DISCUSSION

Building on the results obtained, in this section, we discuss our main findings and provide design recommendations.

Lack of use of assistive technology: All our participants depended on a sighted person to navigate unfamiliar indoor places, regardless of the characteristics of the place (small or big, easy or complex). In contrast to other blind people, as mentioned in Guerreiro et al.'s study [17], who are normally independent, except in some complex areas such as airports due to challenges like walking and carrying luggage, going through security, and moving through areas with a lot of people. We find that one of the main reasons is the lack of use of assistive technologies. Additionally, the nature of Saudi society facilitates the availability of sighted guides. Half of our participants rely on strangers to guide them in unfamiliar places. P2 stated that "sometimes they not only guide us but also ensure that we arrive safely." A future direction could be conducting an ethnographic study to understand the uniqueness of the navigation habits of visually impaired people in the Arab region.

The breadth of spatial mental mapping in a person with visual impairment: All our participants only required assistance from a sighted person during their first visit to an unfamiliar building to create a cognitive map. As the participants described, this map is vast; it starts with a few routes to a very complex map of the stores in a multilevel shopping mall. This differs from the cognitive maps of the participants in Fallah et al.'s study [11]. The authors indicated that their participants have a small cognitive maps, thus they preferred a study in a small area fearing of getting lost. Although all the participants in our study and Fallah et al.'s study used a sighted person when navigating an unfamiliar environment, the breadth of cognitive maps is different. That might be because of the

difference in the involvement of sighted guides [22] and the nature of the participants (e.g., P2 mentioned that “If I am lost during my second visit, I ask someone for help”). The impact of human guides or tools in creating cognitive maps is an area that requires further investigation.

Perceived workload analysis: The perceived workload in the volunteer-based system was less than that in the self-guided system. Based on the NASA-TLX results, the volunteer-based system has less perceived work demand, with the exception of mental demand. The significant difference was in temporal demand (time taken to complete a task), although the exact time for completing the task was not really significant. This indicates that some aspects of the volunteer-based system made the participants feel that they completed it faster, which might include greater trust in people than technology, as we found in the first interviews.

The mental demand was Lower in the self-guided system, based on the difficulty faced when communicating online. Issues that we noted include differences in dialects, such that the participants needed to explain some words. The volunteer also sometimes spoke quickly, which presented a challenge to the participants (P5: “the volunteer in BeMyEye was speaking quickly, that was mentally demanding”).

Errors analysis: There were more errors in the self-guided system than in the volunteer-based system, but the participants were able to self-recover from those errors in the self-guided system. For example, if a participant left the planned route, the system would detect this and ask them to return to the correct path, but the volunteer-based system could not do the same. Participant 4, for example, was notified that she had reached the emergency exit, but it was not an exit, it was an entrance to another part of the building. Such errors cannot be easily recognized or fixed using the volunteer-based system.

Impact of using assistive technology: All the participants were willing to use assistive technology and were aware of the tools’ benefits (P1: “to move independently”). However, it seems that using assistive technology is not easy for them. Besides the high perceived mental workload of using two different tools, participants immediately distrusted a tool after encountering a single error. Most of the available tools in the market are not in Arabic nor designed to be used in Saudi Arabia. There is a need for further research to design and train machine learning models to be used in Saudi Arabia.

We chose two of the three types of the most commonly used navigation tools; one relies on NavCog, similar to mobile navigation, while the other uses volunteers. We excluded the image recognition application, an approach between self-guided and volunteers. Further studies comparing subtle differences between the three conditions, especially in the Arab region, could lead to interesting conclusions.

Design recommendations: Based on our studies, we offer five design recommendations. First, the system should notify the user when the internet connection has been disconnected to enhance their trust and confidence. Second, continuous instructions should be provided to keep the user in touch; this does not mean repeating instructions, but rather responding to the user’s unexpected actions (e.g., when they stop moving). Third, “steps” should be used instead of meters in measuring the path. Fourth, obstacles that might face

blind users should be recognized, such as slopes, moveable objects, and columns. Finally, a voice interface should be personalized and can understand different languages and dialects since the Arabic language is a Semitic language that is widely spoken with many dialects [16].

7 CONCLUSION AND FUTURE WORK

The challenge of navigating unfamiliar indoor buildings is real for blind people. People who live in Arab countries may have their own difficulties and needs. In this research, we investigate the needs and challenges of blind Saudi people. We provide design recommendations to mitigate these challenges and help these individuals navigate easily and confidently using technical assistive technologies.

The future direction of this research includes conducting a long-term study with more participants engaging in more complex navigation tasks (e.g., long, crowded paths) this can help in the collection of data to design and train machine learning models to be used in Saudi Arabia.

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